End-to-end modeling of kilonova light curves and waveforms from neutron star mergers and its possible constraints on nuclear physics parameters

by

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ABSTRACT

This work presents the current efforts to constrain nuclear physics parameters given the precision of up to date telescopes and gravitational wave deterctors. We focus on the equation of state of dense matter and has implications for the waveforms and electromagnetic signals from neutron star mergers. we aim to provide tighter constraints on the symmetry energy and its density dependence. Our analysis includes a comprehensive review of recent observational results, theoretical models, and statistical methods used to extract information about nuclear matter properties from astrophysical data.

Thesis supervisor: Neelima Govind Kelkar
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Acknowledgments

Este trabajo respresenta el final de una de las etapas mas bonitas de mi vida. Llevaré con mucho cariño el recuerdo de todos mis amigos a donde me lleve la motivación.

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Chapter 1

Introduction

One paragraph to tell globally what will be done in this chapter. why etc...

1.1 The messengers

All different kinds of messengers. Update of what have been observed so far, and what we expect to observe in the future.

1.2 Mechanisms

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Gravitational waves

1.2.1 Electromagnetic signals

Gamma rays burst

Thermal emissions

1.3 Summary

Nomenclature for Chapter 1

Roman letters

$\mathcal{C}$	material curve
$\mathbf{r}$	material position [m]
$\mathbf{u}$	velocity [m s^{-1}]

Greek letters

Γ	circulation [m^2 s^{-1}]
ρ	mass density [kg m^{-3}]
$\boldsymbol{\omega}$	vorticity [s^{-1}]